Zoo101

Important MCqs from lectures from 7 to 18

54) Phase contrast microsc qopy first described in……?

1934

55) Who was awarded Nobel Prize in physics in 1953?

Fritz Zernike

56) Phase-contrast microscopy is particularly important in……?

biology.

57) Phase-contrast microscopy is particularly important in biology and It reveals many……?

Cellular structures

58) Phase Contrast Microscope based on the.....?

wavelength (nature) of light rays

59) Different shade of grey are distinguished to our eyes Principle Phase Contrast Microscope due to .....?

Differences in amplitude of light rays.

60)..... converts invisible small phase changes caused by the cell component in to visible intensity changes.

PCM

61) When diffracted and surrounding waves do not over lap.

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Ans: PCM

61) When diffracted and surrounding waves do not over lap, this condition is known as .....?

Answer: Out of phase

62) In a Phase contrast microscope, one set of light rays comes directly from the.....?

Answer: light sources s. waves

63) Light passing through one material & into another material of slightly different refractive index or thickness will undergo a change in.....?

Answer: phase.

64) Phase contrast is obtained with the help of the .....?

Answer: Phase ring

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65) .....is the ratio of Velocity of light in air to that of in given medium.

Answer: Refractive index (μ)

66) How many types of Phase contrast microscopy?

Two main (positive and negative)

67) Light can be considered a form of wave motion consisting of .....?

Answer: sinusoidal waves

68) .....is a technique for obtaining high resolution images of biological and non-biological specimens.

Answer: Electron microscopy (EM)

69) The high resolution of EM images results from the use

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67) Light can be considered a form of wave motion consisting of ……?

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68) ……..is a technique for obtaining high resolution images of biological and non-biological specimens.

Answer: Electron microscopy (EM)

69) The high resolution of EM images results from the use of……?

Answer: electrons

70) ……..provide key information on the structural basis of cell function and of cell disease.

Answer: EM images

71) In ……developed the electromagnetic lens.

1926 HANS BUSCH

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72) The first prototype electron microscope, capable of four-hundred-power magnification, was developed in 1931 by the physicist .....and the electrical engineer .....?

Answer: ERNST RUSKA & MAX KNOLL

73)..... produced the first commercial electron microscope in 1938

Answer: Siemens

74) How many types of electron microscope?

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74) How many types of electron microscope?

Answer: 2 Transmission EM (TEM), Scanning EM (SEM).

75) The very best light microscopes have a resolution limit of about.....?

Answer: 0.2 µm,

76) Viruses are too small to be seen with.....?

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78) The probability of electron sc[illegible] by the..... of the specimen atoms.

Answer: density (atomic number)

79) Two other important techniques for preparing specimens are .....?

Answer: negative staining and shadowing.

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80) In shadowing, a specimen is coated with a thin film of platinum or other heavy metal by evaporation at an angle of about ..... from horizontal?

Answer: 45°

81) The key feature of fluorescent microscopy is that it employs reflected rather than .....?

Answer: transmitted light

82) .....include a set of a compatible excitation filter, emission filter, and dichroic beam splitter; – An excitation filter selects the wavelengths to excite a particular dye within the specimen.

Answer: Optical filters

83) (TIRFM) Stands for?

Answer: Total internal reflection fluorescence microscopy

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84) .....is an optical device that splits a beam of light in two.

Answer: Beam splitter

85) A xenon arc lamp is a highly specialized type of .....?

Answer: GAS discharge lamp,

86) Sigma and pi bonds are chemical.....?

Answer: covalent bonds

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87) Sigma bonds are formed by.....?

Answer: end-to-end overlapping

88) sigma bonds are..... than pi bonds.

Answer: stronger

89) To observe the sample through a fluorescence microscope, it should be first labeled with a fluorescent dyes/substance known as a .....?

Answer: fluorophore.

90) The emission spectrum is usually .....than the excitation spectrum.

Answer: sharper

91) .....is a critical tool for academic and pharmaceutical research, pathology, and clinical medicine.

Answer: Fluorescence microscopy.

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Answer: sharper

91) ……..is a critical tool for academic and pharmaceutical research, pathology, and clinical medicine.

Answer: Fluorescence microscopy.

92) Fluorescence microscope is used to tracing the location of a specific ……..in the specimen

Answer: protein

93) Confocal microscopy was pioneered by …..in 1955.

Answer: Marvin Minsky

94).....is the science in which we have some measurement of the dimensions of an object being observed under the microscope.

Answer: Micrometry

95) How many Types of Micrometers?

Answer: Two (stage micrometer and 2. ocular meter or ocular micrometer)

96) The ocular micrometer is a glass disc with 100 equal divisions or lines on it but with no absolute value

Answer: 100

97) .....the comparison of measurement values delivered by a device under test with those of a calibration standard of known accuracy

Answer: calibration

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